Fiscal Year:	FY 2027	**Task Last Updated:**	FY 06/19/2026
PI Name:	Fischer, Ute Ph.D.		
Project Title:	Supporting Collaboration and Connectedness Between Space and Ground at Lunar Latencies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332-0165	**Congressional District:**	5
Comments:	NOTE: The NSSC also lists the PI as Ute Fischer-Loss (Ed., March 2025).		
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity
Start Date:	03/01/2025	**End Date:**	04/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2026 per L. Juliette/HFBP (Ed., 1/19/26). NOTE: Period of performance (POP) has changed per NASA JSC and NSSC documentation. The new POP is 03/01/2025-02/28/2026. The original POP was: 10/01/2024-09/30/2025.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mosier, Kathleen Ph.D. (Teamscape LLC)		
Grant/Contract No.:	80NSSC25K7247		
Performance Goal No.:			
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Task Description:	Delays in space-ground communication due to signal latency are an inevitable feature of future space exploration missions and may degrade the quality and effectiveness of remote team members' communication. Past work focused on latencies characteristic of Mars missions; however, little is known about the impact of the shorter delays characteristic of lunar exploration. Because communication delays at the Moon will be relatively brief, space crews and ground support personnel may be inclined to rely on voice communication, as was done during the Apollo missions and continues today aboard the International Space Station (ISS). Nevertheless, even short communication delays may pose challenges. A small study of remote collaboration suggests that latencies as brief as three seconds can degrade the quality of voice communication and impair team members' ability to maintain common ground—a shared understanding that is essential for effective communication and a sense of connectedness among remote collaborators. This concern is particularly relevant given evidence that strong social relationships between space crews and mission support personnel on Earth are critical for effective collaboration. The present project sought to examine the viability of voice communication under the low signal latencies characteristic of lunar missions and to evaluate Braiding, a software-based text communication tool, as a potential alternative to voice communication under these conditions. Braiding is designed to support conversations that feel natural despite the presence of communication delays. The software conceptualizes a written conversation between two remote partners as a rotating carousel of multiple "braids" (panels), each dedicated to a distinct topic and moving according to a predetermined rhythm intended to mask the effects of signal latency. By coordinating users' attention and behavior around a shared rhythm, Braiding may foster a sense of copresence—the feeling of being present at the same moment and connected to another person despite physical separation. Building on previous work, the study examined latencies representative of lunar missions and compared Braiding with voice communication to evaluate the extent to which each medium supports remote partners' communication and task performance while also promoting copresence and psychological closeness. As a proxy for space-ground collaboration, the research focused on the Private Medical Conference, a type of operational conversation that has important health implications for crews during long-duration missions. Private Medical Conferences also require organized and coherent communication, involve measurable performance components, and depend on psychological connectedness and the perception of being engaged in a real conversation with a real person. The study design allowed us to determine whether short latencies undermine the viability and benefits of voice communication during Private Medical Conferences relative to Braiding, and to compare subjective measures, such as ratings of communication quality and copresence, as well as objective measures, such as communication effectiveness and task performance, across the two media. The study relates to the expected benefits of Braiding as an effective countermeasure to the risks of communication delays to the functioning of the space/ground team and to crew health during Artemis missions. Specifically, the work will address NASA Human Research Roadmap (HRR) gaps: Team-105 ("Need to identify countermeasures to support team function"), and BMed 106 ("Identify effective strategies to maintain personal relations/interactions to mitigate adverse C/P/Psy/N outcomes during increasingly Earth-independent long duration missions").
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of this study relates to the expected benefits of Braiding as an effective countermeasure to the risks of communication delays to the functioning of the space/ground team and to crew health during Artemis missions.
Task Progress:	The study was conducted online and involved pairs of participants, all flight surgeons or individuals with related expertise, who role-played an astronaut and a flight surgeon during a Private Medical Conference. Six volunteers forming three space-ground teams first participated in a pilot study to provide feedback on the study materials, procedures, and communication tools. An additional twelve volunteers—ten flight surgeons or residents and two medical crew trainers—served as study participants. Participants were paired based on availability and remained together as teams throughout the study. Within each pair, the less experienced participant was assigned the role of astronaut. Each team completed two two-hour test sessions. Participants used voice communication during the first session and Braiding during the second. Prior to participation, short training videos introduced the effects of signal latency on voice communication and demonstrated the use of both the voice communication software and Braiding. The voice session consisted of three Private Medical Conferences (PMCs): one conducted synchronously with no communication delay, one with a five-second one-way delay, and one with a fourteen-second one-way delay. These delays represent the lower and upper bounds of expected communication latencies during lunar missions. To familiarize participants with the procedure, the session began with a practice PMC scenario divided into two ten-minute segments, one conducted with a five-second delay and the other with a fourteen-second delay. The Braiding session included only one PMC because a single Braiding configuration could accommodate both five- and fourteen-second communication delays. However, because Braiding was unfamiliar to participants, they first completed three twenty-minute practice PMCs to gain experience with the system. Nine scenarios involving different spaceflight-related medical issues were developed based on examples reported in the aerospace medicine literature and generated with assistance from ChatGPT. A subject matter expert—a current NASA flight surgeon—and the six pilot participants rated the scenarios for complexity. Based on these evaluations, four moderately complex cases were selected as test scenarios, while four relatively straightforward cases were designated as practice scenarios. Two versions of each scenario were created. The astronaut version provided the health history of a fictional crewmember and described an acute medical issue, including all clinically relevant information necessary for the flight surgeon to reach a diagnosis and develop a treatment plan. The flight surgeon version included only background information about the crewmember, requiring participants to obtain diagnostic information through the PMC. Neither version included information about the diagnosis or treatment plan. Instead, these elements of the case descriptions served as models of ideal solutions and were used to analyze participants' diagnostic processes. Following each PMC, participants completed a brief survey assessing their impressions of the medical issue discussed, the quality of communication, the salience of communication delays, feelings of psychological closeness and copresence, and the mental workload, temporal demands, and frustration experienced during the interaction. Objective

measures focused on two areas. First, the quality of participants' shared task model was assessed by examining the degree to which the astronaut and flight surgeon recalled the same treatment and management plan. Second, task performance was evaluated based on the comprehensiveness of the diagnostic process. Communication analyses focused specifically on disruptions in turn-taking and instances of missing responses.

Overall, the results suggest that both voice communication and Braiding are viable media for supporting operational space-ground interactions under lunar-like communication delays.

Participants reported levels of psychological closeness and copresence during delayed voice PMCs that were comparable to those experienced during synchronous interactions. Similarly, communication delays generally did not increase perceived effort or reduce communication quality. One notable exception emerged: a fourteen-second communication delay significantly lowered participants' ratings of communication quality compared with synchronous communication. However, analyses of the recorded conversations revealed that neither the five-second nor the fourteen-second delay negatively affected the communication process itself, as participants experienced little difficulty coordinating turn-taking.

Participants rated Braiding comparably to delayed voice communication in terms of communication quality, copresence, psychological closeness, mental workload, temporal demand, and frustration. Moreover, the findings suggest that Braiding may improve the efficiency of joint task performance under communication delay. By organizing conversations around distinct topics, the tool appeared to streamline information gathering and decision-making processes. PMCs conducted through Braiding were characterized by more consistent medical check-ins and faster diagnostic processes than delayed voice consultations. Participants tended to address a greater proportion of diagnostically relevant information held by the astronaut, and flight surgeons reached correct diagnoses more quickly.

However, contrary to expectations and the underlying design philosophy of the tool, Braiding did not consistently mask communication delays. Some participants reported that the delays became more noticeable while using the system, particularly as they waited for responses from their partners. Such observations likely reflect the challenge of adapting well-established communication habits and consultation practices to a novel conversational structure.

Despite its unfamiliarity, participants were able to use Braiding successfully and derive measurable benefits after only limited exposure. This finding is encouraging and highlights the value of examining the tool further in longer-term simulations of space missions, where users would have greater opportunities to adapt to its unique communication framework.

Bibliography Type: Description: (Last Updated: 11/18/2025)